Can Local Quantum Rules Reproduce a Dirac Particle?

A Plain-Language Account of a Prospective, Audited Test

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Abstract

This project asked a deliberately limited question. If we place quantum information on a one-dimensional ring and allow it to interact only with nearby neighbors, can two rulebooks, each fixed before its own counted outcomes, reproduce the same carefully chosen part of the behavior of a Dirac particle? Both registered rulebooks received positive development and single-use confirmation classifications, and a separately written high-precision program reconstructed their recorded computations. The surviving evidence covers finite time, low momentum, one particle, declared changes of mass and position-dependent external potential, and a fixed collection of accuracy, causality, symmetry, and resource checks.

A later hostile audit materially narrowed the headline. One empirical monitor for unwanted zone-edge behavior contained a factor-of-two error in both the main and replay programs; it selected no modes. Those empirical values and one related rough control are withdrawn. A different, pre-outcome analytic bound used the correct zone edge and remains far below its frozen threshold. The audit also established that the two free one-particle rules are exactly related by a stroboscopic time-frame change represented by opposite half-cell phases, after remapping the mass angle. They are therefore two separately registered specifications, not proof of two independent architecture classes. The defensible result is qualified constructive sufficiency for the intact shared test core. It is not a derivation of physics from information, a discovery of the Dirac automaton, or a model of an interacting quantum field.

1 The question behind the test

Many physical theories describe change through local rules: what happens here depends on what is already here and nearby, not on an instantaneous command from the other side of the universe. Quantum cellular automata and quantum walks make that idea unusually explicit. Space is represented by a grid, quantum information occupies the grid, and a clock applies a repeating pattern of small local operations.

This study asks whether such a construction can reproduce a bounded *operational role* of a Dirac particle. "Operational role" means a declared bundle of things the construction must do: evolve the right states, respond correctly to specified interventions, produce the right measured distributions and currents, respect causal and symmetry conditions, and stay within fixed error and resource budgets.

The wording matters. We did not ask whether the construction *is* an electron, whether the universe is literally a grid, or whether information is the unique substance of reality. The target Dirac equation, quantum building blocks, lattice, clock, constants, input states, measurements, and permitted

interventions were supplied. What had to be constructed and tested was an explicit local mechanism under those assumptions.

The target was the one-particle Dirac equation in one space dimension on a periodic region of length 16, over a time horizon of 2 in the registered units. Inputs were confined to an 18-dimensional low-momentum sector. This is a small, controlled laboratory for a relativistic quantum role, not a claim about every state, scale, boundary, or duration.

2 Why matching a curve would not be enough

Even this reduced Dirac problem combines several demands. A state has two coupled directional components. Its evolution is quantum and reversible. It has a maximum causal speed. Positive- and negative-energy components can interfere, producing the rapid current oscillation called *Zitterbewegung*. Changing the mass or applying a position-dependent scalar potential should change the evolution in a prescribed way.

A lattice animation can look persuasive while getting some of these relationships wrong. We therefore froze a linked collection of tests rather than judging trajectories by eye. The primary comparison bounded the largest possible evolution error across the entire declared input sector, not just a few attractive example states. Additional witnesses compared position and momentum probabilities, total and local current, state fidelity, and the Zitterbewegung current history. Other checks covered normalization, causal support, parity, time reversal, intervention mapping, and counted circuit resources.

The study also included mass changes and spatial potentials. These matter because a construction that works only when left alone has not reproduced the same operational role as one that responds correctly when the target theory is intervened on. The intervention schedules were exact formulas fixed before their outcomes were opened; they were not fitted site by site after seeing a result.

There is substantial prior work showing that local quantum walks and cellular automata can approach Dirac dynamics, including rigorous observational convergence and the direct one-dimensional Dirac automaton [1, 2]. Zitterbewegung and potential scattering in that direct family are also established [3]. The novelty here is therefore not the existence or discovery of the automaton. It is the prospective comparison protocol: two separately frozen rulebooks, explicit interventions and resources, single-use confirmation, high-precision reconstruction, and a claim boundary that records what later audit did and did not preserve.

3 The two registered rulebooks

The first rulebook was an *ordered product walk*. In each small time step it applied three kinds of local action in a declared order: a phase associated with the scalar potential, an on-site mixing associated with mass, and conditional transport that moved the two directional components to neighboring sites. Repeating these simple pieces approximates the combined Dirac evolution. Because the pieces are applied one after another, their ordering creates a leading finite-step error. The registration therefore expected the certified error to fall by roughly one half whenever the spatial and temporal resolution doubled.

The second rulebook used the established *direct Dirac quantum cellular automaton* (QCA). Its one-particle update was realized locally as an edge operation followed by an on-site routing swap. In free evolution its leading global error was expected to fall by roughly a factor of four under refinement; splitting in a spatial potential restored a leading factor-of-two pattern. The direct registration had its own lattice sizes, thresholds, and resource clock, frozen for that construction rather than copied from the first one.

These descriptions sound architecturally different, and the registered gate partitions really are different. But the later algebraic audit found an exact relationship underneath them. After translating the mass parameter, the free one-particle updates are related by a stroboscopic "time-frame" change, represented mathematically by opposite half-cell phases on the two directional components [4]. Those phases are not themselves registered integer-cell transport layers. In this same one-excitation description, neighboring product gates can also be fused into the direct parameterized form. Consequently, the declared three-layer versus two-layer counts are faithful descriptions of the frozen protocol clocks, but they are not minimal hardware costs. The experiment covers two registered specifications. It does not establish two algebraically independent laws or a broad class of inequivalent architectures.

4 How the test tried to prevent moving the goalposts

The product study completed first. The direct study was then registered as a separate follow-on, so this was not a simultaneous two-arm trial. Within each study, however, the scientific rule, target, states, interventions, measurements, thresholds, numerical precision, and stopping logic were sealed before the counted outcomes.

Each rulebook first faced one indivisible development batch containing four scenarios. Only a committed development pass accepted by the replay could open a fresh confirmation batch. Confirmation contained two new intervention scenarios and could be used only once. The held-out cases were not intended as a large statistical sample. Their purpose was narrower: to check that the unchanged construction and maps still met the full frozen bundle when given new exact schedules.

The main dynamics used 256-bit outward-rounded arithmetic. Rather than reporting a floating-point estimate and hoping rounding was harmless, the calculation carried certified enclosures that had to contain the true numerical answer. A separately written implementation then rebuilt the candidate and target evolutions at 110 decimal digits through different numerical routes. This was an algorithmically separate reconstruction by the same project, not an external replication.

The audit trail includes failed infrastructure attempts. One used a mathematical tool that assumed real rather than complex data. Another created part of the calculation at ordinary machine precision before switching to the required high precision. A third produced a rigorous error bar that became too wide under a spatial potential. Each stopped under its frozen rules before confirmation. Those failures show why plausible decimal outputs are weaker than a successfully reconstructed certified bound.

5 What the recorded results showed

All four counted version-0.4 permanent result records (called "ledgers")—development and confirmation for each rulebook—recorded positive classifications. The product walk covered four development scenarios and two confirmation scenarios, as did the direct QCA. The intact shared checks included

the uniform evolution bound, observable witnesses, causal support, symmetry requirements, resource accounting, and algorithmically separate reconstruction.

For the product walk, the largest certified evolution distances across its scenarios fell from 0.03000726 to about 0.0150 to 0.007496690 as resolution doubled. For the direct rulebook, the largest free or mass-only distances fell from about 0.000560 to 0.000140 to 0.00003498. Its potential-split distances fell from 0.002759767 to about 0.00138 to 0.0006878673. These are upper bounds, so smaller is better. Their refinement patterns matched the separately frozen expectations.

Those numbers should not be used as a race between the rulebooks. The two registrations used different lattice sizes, schedules, gates, and compilation clocks. The meaningful result is that each met its own intact, prospectively justified criteria—excluding the direct empirical zone-edge monitor—and that the detailed position, momentum, current, fidelity, Zitterbewegung, causal, and symmetry checks supported the same bounded role core.

At the finest declared resolutions, the product construction counted 384 serial rounds of local gates for a free run and 448 with an active potential. The direct construction counted 320 free layers, 400 for one potential schedule, and 416 for the other. These totals expose rather than hide the work charged by each registration. Because gate fusion can change them, they do not establish a fundamental efficiency advantage.

6 The audit that changed the headline

The direct QCA has a special issue near the far edge of the lattice momentum zone. The registered low-momentum inputs were kept far away from that region, and the protocol included both an analytic leakage bound and an empirical monitor intended to measure any probability that reached it.

After the result, a hostile-reader audit found a factor-of-two error in the empirical monitor. The intended zone-edge condition began at three quarters of the maximum lattice momentum. The implemented integer test instead placed the boundary beyond every momentum that could exist on the lattice. It therefore selected an empty set and returned zero regardless of the state. The main measurement program, the separately written replay, and one extra stress diagnostic that did not decide pass or fail all copied the same mistaken condition.

All empirical zone-edge probabilities, their pass flags, and the zero reported by that stress diagnostic are withdrawn. They are not repaired in place, and the historical ledgers are not rewritten. An audit-only use of the correct condition makes that rough stress diagnostic nonzero, about 3.49×10^{-8}, confirming that the old zero was vacuous. This was a post-confirmation, non-gating rough potential case, not one of the registered analytic leakage schedules described next.

The analytic leakage calculation was a different route and used the correct zone-edge boundary. Its largest pre-outcome bound was about 1.84×10^{-164} against a frozen threshold of 10^{-24}. That analytic support remains intact. So the defect does not erase the low-band result, but it does mean the complete originally registered direct role was not correctly executed. The surviving conclusion must carry that qualification.

This episode also exposes an important limit of “independent” software. Different algorithms can still share the same misunderstanding of a written specification. Replay was valuable for precision, construction, and numerical separation, but it did not independently check the written meaning of this

predicate. Reconstruction, specification review, and hostile audit solve different problems; a strong workflow needs all three.

7 What survives, and what does not follow

The intact evidence supports the following bounded statement:

> Two separately registered, strictly local quantum update specifications are constructively sufficient for the intact shared core of a declared one-dimensional Dirac role over the tested low-momentum, one-particle, finite-time domain, under the frozen interventions, precision, resources, and verification rules. The direct registration's empirical zone-edge monitor is withdrawn.

"Constructively sufficient" means that explicit mechanisms were built and met the intact stated tests. "Qualified" means that the invalid monitor is excluded and that the claim stops at the supplied domain. "Two specifications" means two prospective records with different declared decompositions and numerical signatures, not two proven architecture classes.

Several stronger conclusions do not follow:

- The study does not derive quantum theory, spacetime, the Dirac equation, or particles from information alone.
- It does not show that the universe is literally a cellular automaton or that information is the unique foundation of nature.
- It does not discover the direct Dirac automaton, its continuum limit, Zitterbewegung, or potential scattering; those are prior results.
- It does not establish accuracy for every momentum, duration, mass, potential, boundary condition, or resolution.
- It does not show that the registered layer and gate counts are minimal or predict a hardware advantage.
- It does not establish necessity, uniqueness, or architecture independence.

8 The spectrum and many-particle boundaries

The low-momentum restriction is scientifically important. Near the opposite edge of the lattice momentum zone, the direct automaton has an exact partner at high quasienergy, the discrete-time analogue of band energy. After removing a global phase, this partner behaves locally like a Dirac mode with the opposite mass sign. Recent work distinguishes such a high-quasienergy *pseudodoubler* from an ordinary extra low-energy particle [5, 6]. It does not invalidate the registered low-band comparison, whose valid analytic bound charges leakage toward that region. It does rule out saying that the same construction is Dirac-equivalent across the whole momentum zone or simply "doubler-free."

The one-particle boundary is equally important. The direct local circuit is a well-defined hard-core qubit automaton—its local subcells do not allow more than one excitation—but its phase when two excitations meet is not the phase of the standard many-fermion extension. The product registration did not freeze the corresponding vacuum and double-occupation phases in a directly comparable way. Because every primary test used one excitation, this study establishes no collision, interacting field, filled-vacuum, particle-creation, or product-versus-direct many-particle claim. Such questions require a new, symmetrically specified prospective experiment; published multiparticle constructions make clear that extra structure is needed [7, 8].

9 Why the result is worth having

The scientific value is not a surprising new lattice formula. It is a worked example of how to ask a construction question without allowing the claim to expand after the result looks encouraging.

The test made the target behavior executable rather than rhetorical. It charged local operations and numerical uncertainty. It separated development from single-use confirmation. It preserved failed verification lineages. It reconstructed the calculation through a second algorithm. Most importantly, when later analysis found both an exact equivalence and a shared reducer bug, the project narrowed the claim instead of protecting the original headline.

That combination leaves a result that is smaller but more durable. The two rulebooks provide disciplined coverage of two frozen implementations of a common bounded role. The exact time-frame relationship explains why that coverage is not architecture-class breadth. The valid analytic leakage bound explains why the low-band result survives. The withdrawn empirical monitor shows why cryptographic fingerprints and replay are valuable but not sufficient.

The appropriate next science is therefore a separate registration: for example, a genuinely distinct local construction defined under an explicit equivalence rule, or a symmetric many-particle test with both many-particle rules fixed in advance. It should not be added retroactively to make the present paper look broader.

10 Conclusion

Within a finite, low-momentum, one-particle domain, two separately registered local quantum rulebooks met the intact shared core of a demanding Dirac test. Their ledgers recorded positive development and single-use confirmation classifications, and a separately implemented high-precision route reproduced the committed computations as implemented. A later audit withdrew an empirical zone-edge monitor shared by both programs while preserving a correctly derived analytic leakage bound. The same audit showed that the free one-particle rules are exactly related by a change of time frame, so they cannot support a claim of independent architecture classes.

The honest conclusion is qualified constructive sufficiency under declared inputs and boundaries. That is less dramatic than deriving reality from information. It is also more useful: it says exactly what was built, exactly what was tested, exactly what broke under audit, and exactly what evidence is left.

Evidence and reproducibility

The technical manuscript, qualified result, post-result audit, generated data, and build scripts are archived together in the project repository. This companion contains no figures, and its numerical result macros are generated from the same content-addressed ledgers used by the technical paper. The repository manuscript README records the exact paths and build command.

References

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